

Evaluation of Li-Mg-B alloy anode for intermediate temperature lithium batteries

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Abstract. A ternary Li-Mg-B alloy (64%Li, 4%Mg and 32%B, all compositions are reported as weight percent in this paper) foil was prepared and its phase structure and discharge behavior were investigated. The main phases existing in the alloy included Li_5B_4 , Li_3Mg_7 , LiB and Li. Coupled with $\text{LiNO}_3\text{-KNO}_3$ eutectic electrolyte and MnO_2 cathode, the anode exhibited great potential in the intermediate temperature lithium batteries. At temperatures above 200°C , two steady voltage plateaus were obtained at near 3V and 2.7V, and the cell displayed an open-circuit voltage exceeding 3.2V. The experimentally measured capacity of the Li-Mg-B alloy was $5427\text{C}\cdot\text{g}^{-1}$ at 200°C , which was higher than that of LiB alloy (70%Li). Adding magnesium to LiB alloy could lower the cost of anodes without capacity loss.

Introduction

Since the LiB anode alloys were discovered by Naval Surface Weapons Center in USA in 1972, various measures to improve their electrochemical performance and to lower their cost have been explored. A.F.Sammells studied the impact of adding some silicon to Li-B alloys and found that discharge EMF of $\text{Li}_{10}\text{Si}_2\text{B}$ was only 30mV higher than LiB, but the capacity was affected much less; $\text{Li}_8\text{Si}_2\text{B}$ provided slightly more capacity than $\text{Li}_{10}\text{Si}_2\text{B}$ but still much less than LiB alloys and the discharge EMF was 137mV higher (all discharge EMF and capacity are reported for the 1st transition) [1-3]. Ternary Li-Mg-B alloys were firstly produced by R.A.Sutula and F.E.Wang in 1975 but no electrochemical tests were conducted [4].

$\text{LiNO}_3\text{-KNO}_3$ eutectic, whose melting point is only 124.5°C , has already been extensively studied as electrolyte materials by many researchers. It was selected here as electrolyte because its melting point was quite compatible with our intended applications. To avoid the serious exothermic reactions between nitrate eutectic electrolytes and sulfide cathodes at elevated temperatures, MnO_2 cathodes were coupled with $\text{LiNO}_3\text{-KNO}_3$ electrolytes to produce single cells. These single cells were tested at a temperature range of from 150°C to 300°C and at a current density of $10\text{mA}\cdot\text{cm}^{-2}$ which was thought enough for most “keep-alive” applications for memory retention in the MWD (measurement while drilling) data logger [5]. The objective of our work was to study the discharge behavior of the Li-Mg-B alloy coupled with low melting point eutectic and evaluate the potential use for intermediate temperature applications ($150\text{-}300^\circ\text{C}$), such as geothermal and oil/gas borehole power sources. This paper will present the results of galvanostatic discharge tests.

Experimental

Materials. The Li-Mg-B alloy was produced by fusing these three simple substances together, the mass percents were 64%, 4% and 32%, respectively. The detailed process can be referred to the preparation of LiB alloys [6, 7]. Anodes were obtained by punching disks (15.5mm diam, 0.08g mass) from Li-Mg-B foil (0.44mm thickness). All salts were vacuum-dried at 130°C overnight before use, the eutectic ratios of electrolytes (33.2%LiNO₃, 66.8% KNO₃) were weighed out and mixed with MgO (35%) and then fused in a quartz crucible at 300°C for 16h. All the reagents with purities over 98.5% were obtained from Sinopharm Chemical Reagent Co, Ltd (SCRC). The catholyte was prepared by blending 70% MnO₂ with 20% electrolyte and 10% graphite (graphite was added necessarily because MnO₂ is not an electronic conductor). Then the mixture was vacuum-dried at 80°C for 8h to eliminate unwanted moisture.

Characterization. To identify the phase structure of Li-Mg-B alloys, XRD was performed on Rigaku D/MAX-2200/PC diffractometer with Cu K α radiation, the scanning range was from 10° to 90° with a scanning speed of 4°/min. To determine the compatibility between Li-Mg-B alloy and LiNO₃-KNO₃ eutectic, simultaneous TGA/DSC was performed on NETZSCH STA 449F3 thermal analyzer. The thermal analysis samples were sealed in an Al₂O₃ pan, the temperature rise rate was 10°C/min and the temperature upper limit was set to 450°C.

Testing of single cells. The single cell was each consisted of a 0.08g Li-Mg-B pellet and a bilayer pellet consisting of 0.50g each of depolarizer-electrolyte (DE) and electrolyte binder (EB) layers pressed at a pressure of 222MPa (4.2 tonnes·cm⁻²). Then the two pellets were sandwiched between stainless steel current collectors with Ni lead wires. After prepared, the single cell was put between platens in a thermostat. Asbestos flakes were fixed between the single cell and the platens as insulators. The discharge measurements were performed on an automatic galvanostatic system (DC-5) in a temperature range of 150°C-300°C. A steady-state load of 19mA (10mA·cm⁻²) was applied which was consistent with the actual need of the drilling devices. The discharge process was terminated when the voltage dropped below 1.00V. All preparations, processing and handing with materials were conducted in a glove box under an atmosphere of high-purity argon (<1ppm of water and oxygen).

Results and discussion

Phase structures of the Li-Mg-B alloy. The glass substrate used to hold the XRD samples must be covered with a plastic film because lithium is quite easy to react with moisture in air. The XRD file of the Li-Mg-B alloys is illustrated in Fig.1.

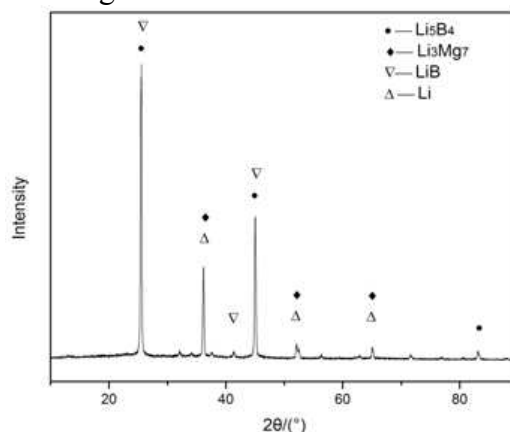


Figure 1. XRD profile of Li-Mg-B (64%Li, 4%Mg and 32%B) sample

Several peaks of different phases match together as can be seen from Fig.1, so it is hard to conclude the exact phase structure of Li-Mg-B alloys only by XRD method. The possible main phases identified from the sample are Li₅B₄, Li₃Mg₇, LiB and Li, which are similar to those in early reports

on LiB alloys except Li_3Mg_7 [8, 9]. Though a clear understanding of the structure of LiB alloy has not been reached completely, there is a consensus that both Li and Li alloys exist and Li alloys can hold liquid Li as a porous matrix even above 1000°C [10]. The result of our work is consistent with this basic knowledge. In addition, a new compound: Li_3Mg_7 , appears as a result of the addition of Mg. Due to the limited content of Mg, a binary solid solution is most likely formed which can improve the melting point of Li and inhibit the flow of liquid Li at high temperatures.

Thermal analyses. It is necessary to accomplish thermal analysis prior to preparation and testing of single cells to avoid any risks. The compatibility of the Li-Mg-B alloy with the $\text{LiNO}_3\text{-KNO}_3$ eutectic was examined by simultaneous TGA/DSC analyses. The results are shown in Fig.2.

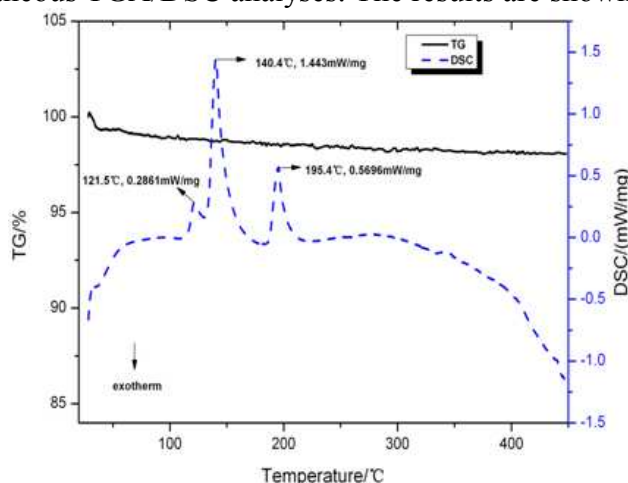


Figure 2. Simultaneous TGA/DSC responses of the Li-Mg-B alloy mixed with $\text{LiNO}_3\text{-KNO}_3$ eutectic

In the temperature range applied from room temperature to 450°C , there is no obvious exotherm but only three endotherms found from the DSC curve. The sharpest peak at 140.4°C may be in agreement with the melting point data of the $\text{LiNO}_3\text{-KNO}_3$ eutectic (124.5°C) published in the literature but a little higher [11, 12]. Another sharp peak at 195.4°C provides a proof of melting of Li whose melting point is 182.5°C . The phenomenon supports the existence of Li in alloys from one side as a supplement of XRD results. Since LiB anodes often deflagrate in molten nitrate electrolytes at temperatures above 350°C [13] and this phenomenon was not found in Li-Mg-B alloy below 450°C , so addition of Mg to LiB alloys can improve thermal and chemical stabilization up to 450°C at least. TGA examination showed that the mixture was thermally stable up to 450°C . The slight loss of the total mass at the outset may correspond to the elimination of moisture in the mixture. All the results proved no adverse exothermic reactions occurred during the temperature range conceived for the applications. In other words, it is quite safe for the chemistry combinations designed.

As other researchers reported, the only reason for the security of the designed cells is a passive Li_2O film which is one of the reaction products between Li alloys and nitrate electrolytes [14, 15]. The reaction is shown in Eq. (1). Though violent chemical reactions may occur at elevated temperatures due to breakage of Li_2O films, the thermal analysis results insure that the chemistry can work safely at $150\text{-}300^\circ\text{C}$.



Discharge behavior of single cells. Discharge traces for Li-Mg-B/ $\text{LiNO}_3\text{-KNO}_3\text{/MnO}_2$ cells are shown in Fig.3 as a function of capacity. There is no considerable voltage plateau found at 150°C , the reason for this phenomenon results from the limit of kinetics of this chemistry at lower temperature close to the melting point of eutectics. The short voltage plateau and linearly declining voltage are both not adapted to the actual applications. However, with the Li (Al) or Li (Si) alloys as the anodes, much better discharge behaviors at 150°C were attained by Ronald A.Guidotti etc [16, 17]. At

elevated temperatures of 200-300°C, two distinct voltage plateaus all exist at near 3V and 2.7V which are characteristics of multiphase discharge redox processes. The total discharge capacities are all above 85mAh, the maximum of 109.50 mAh comes forth at 200°C with the highest voltage at the second plateau.

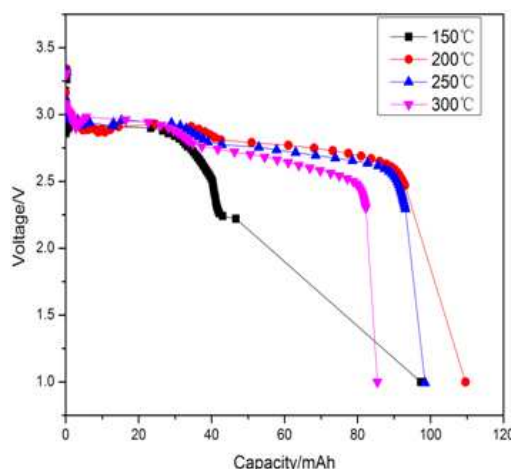


Figure 3. Effect of temperature on the discharge behavior of Li-Mg-B/LiNO₃-KNO₃/MnO₂ single cells at 10mA·cm⁻²

In order to understand the specific reaction mechanism, complete discharge tests with a cut-off voltage of 0V were performed at 200°C on single cells consisting of 0.50g or 1.00g DE respectively, other conditions consistent with the above. The results are presented in Fig.4. For the single cell consisting of 0.50g DE, a total capacity of 109.70 mAh was released and the corresponding data for the single cell consisting of 1.00g DE was 120.60 mAh. As Ronald described [16], the following reaction-Eq. (2) may be responsible for the observation.



According to the reaction, capacity of 107.72 mAh can be obtained from 0.35g MnO₂ which exists in 0.50g depolarizer-electrolyte pellet. It is quite close to the experimental value which shows that the former was designed as cathode limited. To obtain the discharge capacity of the Li-Mg-B alloy, the cathode was doubled to make sure that the single cell was anode limited. Capacity of 120.6mAh was released by 0.08g anode, then it can be calculated that the discharge capacity of per mass Li-Mg-B alloy was 1507.5mAh·g⁻¹ (5427 C·g⁻¹). However, the capacity of Li-B alloys (70%Li) is only 4800 C·g⁻¹ [18], so it can be concluded that by the use of Mg as an additive to Li-B alloys, higher actual discharge capacity and lower cost are obtained since Mg is much cheaper than Li. XRD analysis was performed on DE discharged to identify the intermediate phase, the result in Fig.5 shows two main phases existed as reactions of MnO₂. So the intermediate phase is most likely to be Li₂MnO₃ and the two voltage plateaus on the curve can be explained by the following Eq. (3).



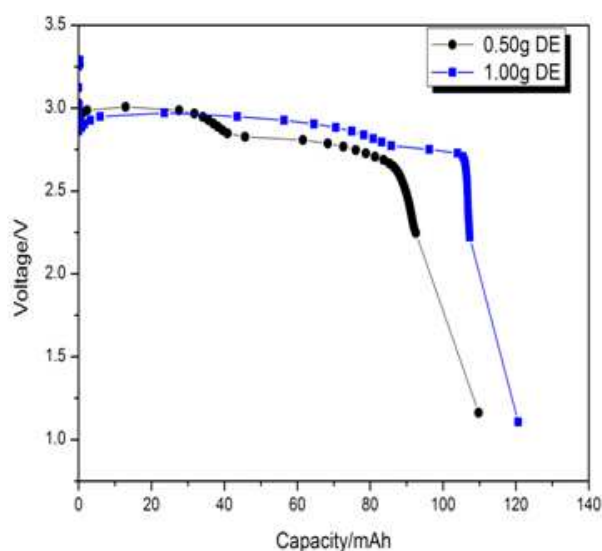


Figure 4. Discharge curves of single cells consisting of 0.50g or 1.00g DE

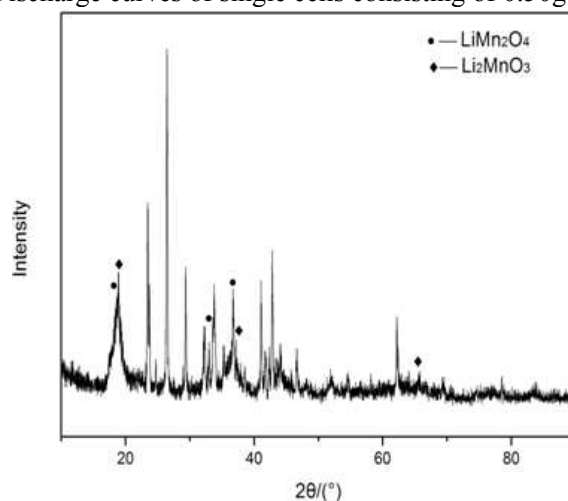


Figure 5. XRD profile of the DE sample after discharged

The capacity and open circuit voltage (OCV) as a function of temperature are summarized in Fig.6. All the cells displayed OCV values of higher than 3.26V which helps to improve specific energy. The transition trend of OCV is quite similar to that of the capacity. The comprehensive performance reaches the best at 200°C and then decays gradually with increasing temperature which may be attributed to increased self-discharge reactions.

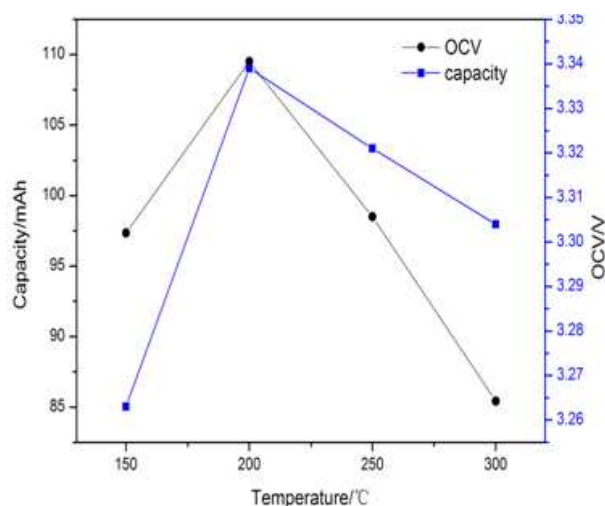


Figure 6. Effect of temperature on the capacity and OCV at $10\text{mA}\cdot\text{cm}^{-2}$

Conclusions

The structure and discharge performance of Li-Mg-B alloy anode were studied: (1) The main phases identified in the Li-Mg-B alloy are Li_5B_4 , Li_3Mg_7 , LiB and Li, which are similar to those in early reports on LiB alloys except Li_3Mg_7 . Since there is still no unanimous understanding of the structure of LiB alloys, the accurate structure of Li-Mg-B also deserves more in-depth studies, which will explain the reserve and discharge mechanism of active Li in Li-Mg-B alloys. (2) The thermal analysis results show that no violent exothermic reactions occur below 450°C , so it is safe for applications. (3) The electrochemical tests show that Li-Mg-B alloy anodes have a high capacity and open circuit voltage. The working life of a single cell can reach near 6h with a cut-off voltage of 1.00V. So the Li-Mg-B cells are competent for the oil/gas or geothermal drilling applications, also can lower the cost of anode materials without any loss of discharge performance. More work about this new anode should be done to evaluate the feasibility of this technology, for example, the effect of changing Mg contents and to get longer working life, batteries consisting of several single cells should be tested in future.

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